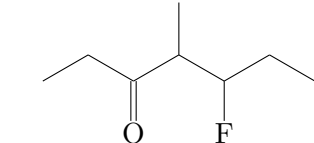
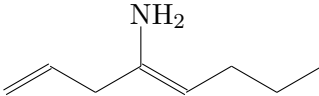
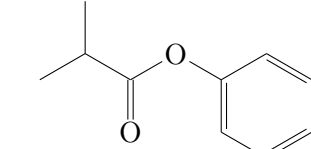


Unit 5: Organic Chemistry Half-Unit Test A

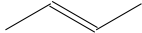
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1. [6 marks] Identify the type of organic molecule(s) based on the functional group(s) in the following:

Structure	Type of Organic Molecule(s) Based on Functional Group(s)
 5-fluoro-4-methyl-3-heptanone	
 4-amino-1,5-octadiene	
 phenyl 2-methylpropanoate	

2. [2 marks] What effect does a double or triple bond have on the density of a compound? Explain.

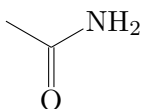
3. [2 marks] Which of the following structures should have the highest MP/BP? Explain.

		
molecule X	molecule Y	molecule Z

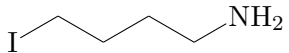
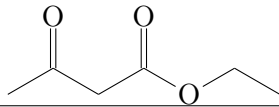
4. What is the chemical formula of a structure that satisfies the following conditions? Show your work analytically as opposed to drawing a structure that fits the description.

a. [2 marks] An aliphatic that has 40 carbon atoms, two cyclic structures & has one triple bond.

b. [2 marks] An alcohol that has 20 carbon atoms, two hydroxyl groups & two double bonds. No cyclic structure.

5. [2 marks] Would adding ethanamide  to water cause boiling point elevation? Explain.

6. [2 marks] Which of the following structures should have the best solubility in water? Explain.

CH₃OH		
molecule X	molecule Y	molecule Z

7. [3 marks] Draw any 3 compounds that are structural isomers of the chemical formula C_5H_8O . Do not name the structures.

8. [2 marks] Why do carboxylic acids tend to be solids or liquids at room temperature?

9. [2 marks] Would methylcyclopropane  be stable at room temperature? Explain.

10. [2 marks] Why do same mass aldehydes & ketones have similar but slightly different MP/BP temperatures? Explain.

11. [1 mark] Draw an example of a 3° amine.

12. [2 marks] Provide an explanation for the following BP data. Be specific.

Molecule	BP Temperature ($^\circ\text{C}$)
$\text{CH}_3\text{-F}$	-78.4
$\text{CH}_3\text{-Cl}$	-24.2
$\text{CH}_3\text{-Br}$	3.56
$\text{CH}_3\text{-I}$	42.43